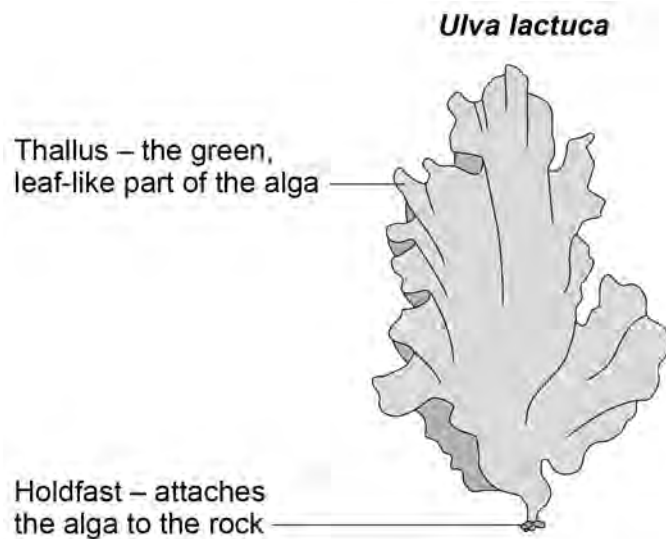


0 7

Ulva lactuca is an alga that lives on rocks on the seashore. It is regularly covered by seawater.

Figure 6 shows a diagram of one *Ulva lactuca* alga.

Figure 6



0 7

1

Unlike plants, *Ulva lactuca* does not have xylem tissue.

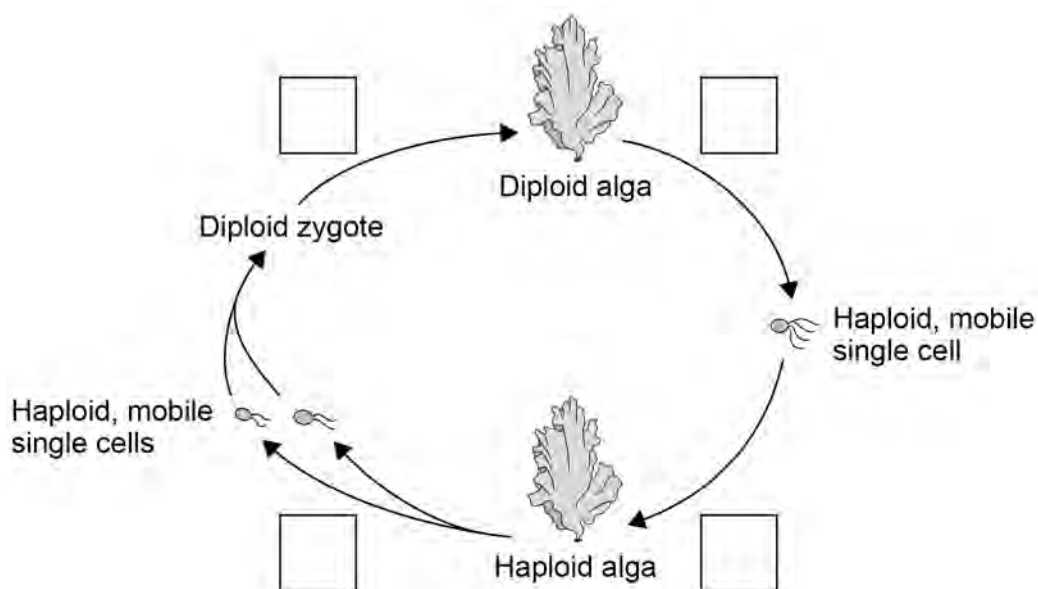
Suggest how *Ulva lactuca* is able to survive without xylem tissue.

[1 mark]

Ulva lactuca has a haploid and a diploid form.

Figure 7 shows the life cycle of *Ulva lactuca*.

Figure 7



- 0 7 . 2** On **Figure 7** complete each box with an appropriate letter to show the type of cell division happening between each stage in the life cycle. Use 'T' to represent mitosis and 'E' to represent meiosis.

[2 marks]

- 0 7 . 3** *Ulva prolifera* also produces haploid, mobile single cells that can fuse to form a zygote.

Suggest and explain **one** reason why successful reproduction between *Ulva prolifera* and *Ulva lactuca* does **not** happen.

[2 marks]
